

BENGHAZI, Libyan Arab JamahiriyaWMO: **620530**Lat: **32.097N**Lon: **20.269E**Elev: **433**StdP: **14.47**Time Zone: **2.00 (E02)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.4	46.2	25.1	19.5	74.5	29.9	24.3	73.9	46.2	59.5	39.3	59.4	5.4	120	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.1	98.8	68.9	95.4	69.0	92.9	68.6	76.9	85.4	75.7	84.4	74.6	83.5	15.7	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.7	132.2	80.5	73.3	126.0	80.2	72.0	120.1	79.2	40.5	85.0	39.4	84.6	38.4	83.4	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	35.7	31.9	28.9	40.7	106.7	2.0	3.8	39.3	109.4	38.1	111.7	37.0	113.8	35.5	116.6	
(5)				39.0	80.2	1.5	3.6	37.9	82.8	37.1	84.9	36.2	87.0	35.2	89.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.0	55.9	56.1	60.2	66.5	73.6	78.0	79.7	81.1	79.6	72.9	65.3	58.6
	DBStd	10.56	3.14	4.17	5.93	7.19	7.57	5.89	3.30	2.85	4.78	4.84	4.63	3.98
	HDD50	3	1	2	0	0	0	0	0	0	0	0	0	0
	HDD65	1039	283	253	176	63	8	0	0	0	0	2	49	205
	CDD50	6945	183	172	317	496	732	841	920	963	887	709	458	267
	CDD65	2504	0	3	28	109	275	391	455	498	437	246	56	6
	CDH74	24290	1	18	235	1044	2832	4088	4653	5281	4295	1585	238	20
	CDH80	10222	0	3	58	386	1352	1932	1922	2251	1825	446	45	2
Wind	WSAvg	13.1	11.8	12.5	13.2	15.1	14.7	14.4	14.3	13.5	12.3	11.4	11.4	12.0
Precipitation	PrecAvg	10.0	2.5	1.6	1.1	0.2	0.2	0.0	0.0	0.0	0.1	0.9	1.1	2.5
	PrecMax	14.5	8.3	5.3	4.4	1.5	1.0	0.3	1.0	0.2	0.7	3.1	2.6	6.3
	PrecMin	5.6	0.9	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.9	1.4	1.2	0.9	0.3	0.2	0.1	0.2	0.0	0.2	0.9	0.8	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.1	76.8	86.1	94.7	102.2	105.4	99.0	98.5	100.6	93.4	85.9	76.7
		MCWB	53.5	56.9	59.8	64.3	71.0	69.7	70.5	70.9	70.8	68.6	61.3	56.7
	2%	DB	66.4	71.0	79.6	89.3	96.5	98.6	95.1	94.7	96.4	88.4	79.7	71.2
		MCWB	54.6	55.6	57.9	61.8	67.7	68.2	70.3	71.7	70.3	66.4	61.1	57.5
	5%	DB	64.4	66.7	75.2	84.4	91.7	93.9	91.4	91.6	92.2	84.6	76.6	68.1
		MCWB	54.5	54.7	56.9	60.7	64.7	67.4	70.4	72.2	69.5	66.2	61.2	57.1
	10%	DB	62.6	64.2	71.4	80.3	87.6	89.9	88.2	89.4	88.7	81.8	73.6	65.9
		MCWB	54.1	54.2	56.1	59.1	63.8	66.6	71.0	72.2	69.6	66.0	60.8	56.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.9	60.7	65.5	72.7	75.6	77.5	77.5	79.3	78.1	74.4	69.0	62.9
		MCDB	64.3	70.1	76.3	85.6	93.6	93.1	86.3	87.9	88.4	82.4	76.4	71.3
	2%	WB	57.2	58.2	61.5	65.7	70.7	73.6	75.7	77.0	75.8	71.5	65.8	60.2
		MCDB	63.1	66.5	72.0	81.3	87.2	84.6	83.9	84.7	85.3	80.1	73.0	66.6
	5%	WB	56.0	56.6	59.2	62.7	68.1	71.9	74.6	76.0	74.3	69.9	64.0	58.8
		MCDB	61.8	63.9	68.6	77.1	83.6	82.3	82.9	84.0	83.9	79.0	71.3	65.4
	10%	WB	54.9	55.3	57.7	60.8	66.1	70.5	73.7	74.9	72.8	68.5	62.4	57.5
		MCDB	60.7	61.8	67.5	74.1	79.5	81.6	82.0	83.3	82.4	77.3	70.1	63.9
Mean Daily Temperature Range	5% DB	MDBR	13.6	14.5	17.5	19.6	19.8	19.4	17.1	17.1	17.7	16.3	15.4	13.2
		MCDBR	15.7	18.6	23.2	25.2	24.7	24.4	22.2	20.6	21.7	19.2	18.0	16.0
		MCWBR	9.3	10.0	10.7	11.5	10.9	11.2	10.0	9.4	9.8	8.9	9.2	9.2
	5% WB	MCDBR	13.9	16.2	19.4	22.4	21.7	19.9	17.6	17.1	17.1	16.1	15.7	14.4
		MCWBR	9.4	10.4	11.2	11.8	11.4	10.3	8.3	8.5	9.9	9.2	10.0	9.6
Clear-Sky Solar Irradiance	taub	0.368	0.400	0.449	0.498	0.486	0.435	0.426	0.429	0.456	0.424	0.389	0.374	
	taud	2.292	2.171	2.015	1.902	1.968	2.163	2.206	2.212	2.125	2.207	2.262	2.290	
	Ebn at Noon	266	268	263	254	258	271	272	271	257	257	256	256	
	Edh at Noon	34	42	53	62	58	48	46	45	47	40	35	33	
All-Sky Solar Radiation	RadAvg	987	1241	1722	2069	2313	2530	2531	2302	1906	1451	1086	872	
	RadStd	49	80	80	79	100	71	35	37	49	64	49	43	